

Alejandro Sánchez Yalí

AI Engineer



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SUMMARY

AI Engineer and mathematician with 8+ years building production software and 3+ years shipping LLM-powered systems end to end. M.Sc. in Mathematics and 11 years teaching mathematics and physics at university, including machine learning in the last year and a half. Designs, builds and deploys multi-tenant AI products: RAG pipelines, multi-provider LLM gateways with fallback, agent orchestration, and real-time conversational systems – owning the architecture as well as the code.

TECHNICAL SKILLS

- **AI & LLMs:** LangChain, LangGraph, LiteLLM, RAG, MCP, OpenAI API, Whisper, ElevenLabs, agentic development with Claude Code
- **ML & Mathematics:** PyTorch, TensorFlow, Scikit-learn, JAX, Flax, NumPy, Pandas, linear algebra, probability, statistical inference
- **Languages:** Python (8+ years), TypeScript (6+ years), JavaScript (6+ years), SQL, Rust (personal projects)
- **Backend:** FastAPI, Django, Celery, Flask, NestJS, Node.js, RESTful APIs, microservices, event-driven design
- **Data & Infrastructure:** PostgreSQL, Redis, MongoDB, Docker, AWS, GCP, GitHub Actions, CI/CD
- **Architecture:** Clean architecture, DDD, SOLID, modular monolith, screaming architecture

EXPERIENCE

Independent AI Consultant

Freelance

Oct 2026 – Present

Remote

- Lead engineer of Aluna, an AI recruitment platform for staffing agencies (see below)

AI Specialist

Lapzo

Aug 2025 – Sep 2026

Remote, Mexico

- Built an HR agent (LangGraph, FastAPI, PostgreSQL) that answers company-policy questions with citations and executes HR requests with human confirmation, with per-tenant isolation, delivered with a runbook and a handover
- Migrated the course-vectorization system to a dedicated VM and moved the embeddings pipeline from n8n into the platform; hardened the LLM course assistant (scoped context, consistent recommendations)
- Designed and built a Digital Professor that holds real-time conversations with learners and explains course content through interactive slides (LLM services, ElevenLabs voice, real-time layer)
- Closed 57 tasks and 6 epics in the first nine months of 2026 with a single production bug, per the team's board
- Built the audit system for courses and learning paths and an AI-powered educational content generator with LangChain, LangGraph and n8n, with Celery workers running the pipelines in the background

Independent Software & AI Consultant

Freelance

Aug 2024 – Jul 2025

Remote

- Lead engineer of CREARIA's multi-tenant WhatsApp and email AI agent platform, in production with several client businesses including a real-estate agency: led a team of three across 5 FastAPI services and 2 Next.js dashboards on AWS ECS, with RAG over pgvector, multi-provider routing through LiteLLM, AI lead scoring and human escalation with full context; also led the AI coaching agent CREARIA delivers to one of its clients
- Designed database architecture for a client's workflow automation platform and served as code reviewer for their development team
- Ran technical design sessions with clients, translating business requirements into system architectures and naming the trade-off behind each technology decision

Full Stack Engineer

Monadical

Apr 2021 – Aug 2024

Remote, Canada/US

- Built VC Tools, an LLM tool that retrieves and processes company information for investors (OpenAI, LangChain)

- Full-stack engineer on ZoHuddle, a video conferencing app for scientific conferences (Neuromatch) that matches participants by mutual interests: React front end and Django back end, with Django Channels (WebSockets) and Twilio Video
- Built and scaled microservices for AI/ML and data-intensive applications; designed ETL pipelines and custom Django ORM extensions
- Did code reviews on a distributed team across Canada, the US and Latin America

Full Stack Engineer

BCFort

Aug 2018 – Oct 2020

Medellín, Colombia

- Developed blockchain and analytics platforms with Python (Django), Node.js, React and Web3.js on AWS, GCP and IBM Cloud

Professor of Mathematics & Physics (part-time)

Universidad de Antioquia

Jan 2010 – Oct 2021

Medellín, Colombia

- Taught mathematics and physics; in the last year and a half also machine learning and data science with Python, NumPy, Pandas, TensorFlow, PyTorch and Scikit-learn

FEATURED PROJECTS

- **Plixiq – AI Customer Support on WhatsApp**  – *Python, FastAPI, LiteLLM, PostgreSQL, Redis, Next.js, WebSocket*
 - Architect and lead engineer of a multi-tenant SaaS where AI agents answer routine customer inquiries, book appointments and escalate the rest to human operators with full conversation context
 - Modular monolith (14 components plus a shared kernel) with boundaries enforced in CI; LiteLLM layer with a primary and fallback provider per agent, so an outage becomes a fallback
- **Aluna – AI Recruitment Platform for Staffing Agencies**  – *Next.js 15, FastAPI, LiteLLM, PostgreSQL, Inngest, Claude, Gemini, WhatsApp Cloud API*
 - Lead engineer of an AI recruitment platform for staffing agencies – 336 source files across 4 packages, 26 tables and 11 external providers – split into a Next.js 15 application owning all domain state and a stateless FastAPI service owning model calls and the WhatsApp channel
 - Durable Inngest analysis pipeline (9 checkpointed steps, 3 retries) where deduplication runs before quota enforcement so a re-analysed CV reuses its cached score instead of paying for another model call; one screening engine serves WhatsApp, web chat and email
- **Digital Human – Interactive AI Avatar**  – *React, Three.js, React Three Fiber, OpenAI GPT, Whisper, ElevenLabs*
 - Open-source conversational avatar with 464 stars and 110 forks: real-time speech-to-text, generation, voice synthesis and lip synchronisation
 - The engineering problem was latency budgeting across four services, since a human notices any pause in a conversation

OPEN SOURCE

- **talking-avatar-with-ai**  – 464 stars, 110 forks. Open-source conversational AI avatar: OpenAI GPT, Whisper, ElevenLabs, Rhubarb lip-sync, Three.js
- **Monadical-SAS/Morpheus**  – 19 pull requests. Open-source AI image generation platform: FastAPI, React, Stable Diffusion

CONFERENCE TALKS

- **Scaling Learning Models: Parallelism Strategies in JAX with Flax**  – PyCon Colombia, 2024

TECHNICAL WRITING

- **Pre-training: the Idea Behind Every LLM** (2026)

EDUCATION

M.Sc. in Mathematics

Universidad de Antioquia

Aug 2010 - Sep 2013

Medellín, Colombia

B.Sc. in Teaching Mathematics and Physics

Universidad de Antioquia

Jan 2004 - Oct 2009

Medellín, Colombia